

Лабораторная работа №5

Администрирование локальных сетей

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11 марта 2025

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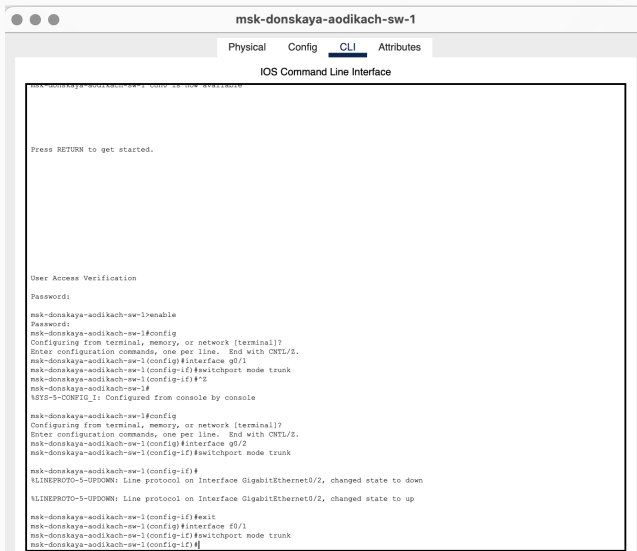
Информация

- Дикач Анна Олеговна
- НПИбд-01-22 (1132222009)
- Российский университет дружбы народов
- <https://github.com/chelibos?tab=repositories>

Получить основные навыки по настройке VLAN на коммутаторах сети.

Выполнение лабораторной работы

Используя последовательность из текста лабораторной работы провожу конфигурацию Trunk-порта на коммутатору msk-donskaya-aodikach-sw-01



```
msk-donskaya-aodikach-sw-1
Physical  Config  CLI  Attributes
IOS Command Line Interface

msk-donskaya-aodikach-sw-1 Come to the console

Press RETURN to get started.

User Access Verification

Password:

msk-donskaya-aodikach-sw-1>enable
Password:
msk-donskaya-aodikach-sw-1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-sw-1(config)#interface g0/1
msk-donskaya-aodikach-sw-1(config-if)#switchport mode trunk
msk-donskaya-aodikach-sw-1(config-if)#%
%SYS-5-CONFIG_I: Configured from console by console

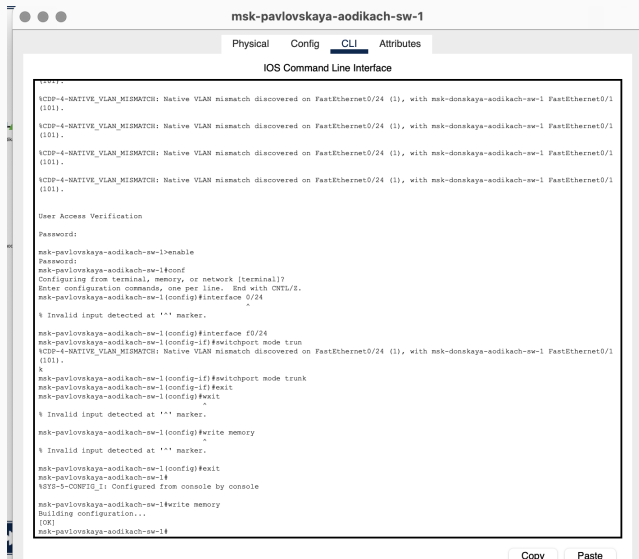
msk-donskaya-aodikach-sw-1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-sw-1(config)#interface g0/2
msk-donskaya-aodikach-sw-1(config-if)#switchport mode trunk

msk-donskaya-aodikach-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

msk-donskaya-aodikach-sw-1(config-if)#exit
msk-donskaya-aodikach-sw-1(config)#interface #0/1
msk-donskaya-aodikach-sw-1(config-if)#switchport mode trunk
msk-donskaya-aodikach-sw-1(config-if)#
```

Используя последовательность из текста лабораторной работы провожу конфигурацию Trunk-порта на коммутатору msk-pavlovskaya-aodikach-sw-1



```
msk-pavlovskaya-aodikach-sw-1
Physical Config CLI Attributes
IOS Command Line Interface

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (1), with msk-donskaya-aodikach-sw-1 FastEthernet0/1 (101).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (1), with msk-donskaya-aodikach-sw-1 FastEthernet0/1 (101).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (1), with msk-donskaya-aodikach-sw-1 FastEthernet0/1 (101).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (1), with msk-donskaya-aodikach-sw-1 FastEthernet0/1 (101).

User Access Verification
Password:
msk-pavlovskaya-aodikach-sw-1>enable
Password:
msk-pavlovskaya-aodikach-sw-1#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-pavlovskaya-aodikach-sw-1(config)#interface 0/24
^
% Invalid input detected at '^' marker.

msk-pavlovskaya-aodikach-sw-1(config)#interface f0/24
msk-pavlovskaya-aodikach-sw-1(config-if)#switchport mode trunk
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (1), with msk-donskaya-aodikach-sw-1 FastEthernet0/1 (101).
X
msk-pavlovskaya-aodikach-sw-1(config-if)#switchport mode trunk
msk-pavlovskaya-aodikach-sw-1(config-if)#exit
msk-pavlovskaya-aodikach-sw-1(config)#exit
^
% Invalid input detected at '^' marker.

msk-pavlovskaya-aodikach-sw-1(config)#write memory
^
% Invalid input detected at '^' marker.

msk-pavlovskaya-aodikach-sw-1(config)#exit
msk-pavlovskaya-aodikach-sw-1#
SYS5-S-CONFIG_1: Configured from console by console

msk-pavlovskaya-aodikach-sw-1#write memory
Building configuration...
[OK]
msk-pavlovskaya-aodikach-sw-1#
```

Прописываю конфигурацию диапазонов портов и конфигурации VTP msk-donskaya-aodikach-sw-01

```
msk-donskaya-aodikach-sw-1
Physical Config CLI Attributes
IOS Command Line Interface
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-sw-1(config)#vtp mode server
Device mode already VTP SERVER.
msk-donskaya-aodikach-sw-1(config)#vtp domain donskaya
Changing VTP domain name from NULL to donskaya
msk-donskaya-aodikach-sw-1(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-aodikach-sw-1(config)#vlan 2
^
% Invalid input detected at '^' marker.
msk-donskaya-aodikach-sw-1(config)#name management
^
% Invalid input detected at '^' marker.
msk-donskaya-aodikach-sw-1(config)#vlan 2
msk-donskaya-aodikach-sw-1(config-vlan)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up
msk-donskaya-aodikach-sw-1(config-vlan)#name management
msk-donskaya-aodikach-sw-1(config-vlan)#vlan 3
msk-donskaya-aodikach-sw-1(config-vlan)#name servers
msk-donskaya-aodikach-sw-1(config-vlan)#vlan 101
msk-donskaya-aodikach-sw-1(config-vlan)#name dk
msk-donskaya-aodikach-sw-1(config-vlan)#vlan 102
msk-donskaya-aodikach-sw-1(config-vlan)#name departments
msk-donskaya-aodikach-sw-1(config-vlan)#vlan 103
msk-donskaya-aodikach-sw-1(config-vlan)#name ads
msk-donskaya-aodikach-sw-1(config-vlan)#vlan 104
msk-donskaya-aodikach-sw-1(config-vlan)#name other
msk-donskaya-aodikach-sw-1(config-vlan)#
^
% Invalid input detected at '^' marker.
msk-donskaya-aodikach-sw-1(config-vlan)#sh vlan
^
% Invalid input detected at '^' marker.
msk-donskaya-aodikach-sw-1(config-vlan)#exit
msk-donskaya-aodikach-sw-1(config)#exit
msk-donskaya-aodikach-sw-1#
$SYS-5-CONFIG_I: Configured from console by console
msk-donskaya-aodikach-sw-1#sh vlan

VLAN Name                Status    Ports
-----
1  default                active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                           Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24
2  management             active
3  servers                active
101 dk                   active
102 departments          active
103 ads                  active
104 other                active
1002 fddi-default         active
1003 token-ring-default   active
1004 fddinet-default      active
```

IOS Command Line Interface

```

3 servers active
101 dk active
102 departments active
103 adm active
104 other active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active

VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrgdMode Trans1 Trans2
-----
1 enet 100001 1500 - - - - - 0 0
2 enet 100002 1500 - - - - - 0 0
3 enet 100003 1500 - - - - - 0 0
101 enet 100101 1500 - - - - - 0 0
102 enet 100102 1500 - - - - - 0 0
103 enet 100103 1500 - - - - - 0 0
104 enet 100104 1500 - - - - - 0 0
1002 fddi 101002 1500 - - - - - 0 0
1003 tr 101003 1500 - - - - - 0 0
1004 fddnet 101004 1500 - - - lewse - 0 0
1005 trnet 101005 1500 - - - lbe - 0 0

VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrgdMode Trans1 Trans2
-----

Remote SPAN VLANs
-----

msk-donskaya-aodikach-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-aodikach-sw-1#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-sw-1(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-aodikach-sw-1(config)#interface range f0/1 - 5
msk-donskaya-aodikach-sw-1(config-if-range)#switchport mode access
msk-donskaya-aodikach-sw-1(config-if-range)#SPANFRES-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/1 VLAN1.
%SPANFRES-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/1 on VLAN0001. Inconsistent port type.

msk-donskaya-aodikach-sw-1(config-if-range)#interface range f0/1 - 15
msk-donskaya-aodikach-sw-1(config-if-range)#switchport mode access
msk-donskaya-aodikach-sw-1(config-if-range)#switchport access vlan 101
msk-donskaya-aodikach-sw-1(config-if-range)#interface range f0/20
msk-donskaya-aodikach-sw-1(config-if-range)#interface range f0/1 - 15
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (101), with msk-pavlovskaya-aodikach-sw
msk-donskaya-aodikach-sw-1(config-if-range)#interface range f0/20
msk-donskaya-aodikach-sw-1(config-if-range)#switchport mode access
msk-donskaya-aodikach-sw-1(config-if-range)#switchport access vlan 104
msk-donskaya-aodikach-sw-1(config-if-range)#exit
msk-donskaya-aodikach-sw-1(config)#exit
msk-donskaya-aodikach-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

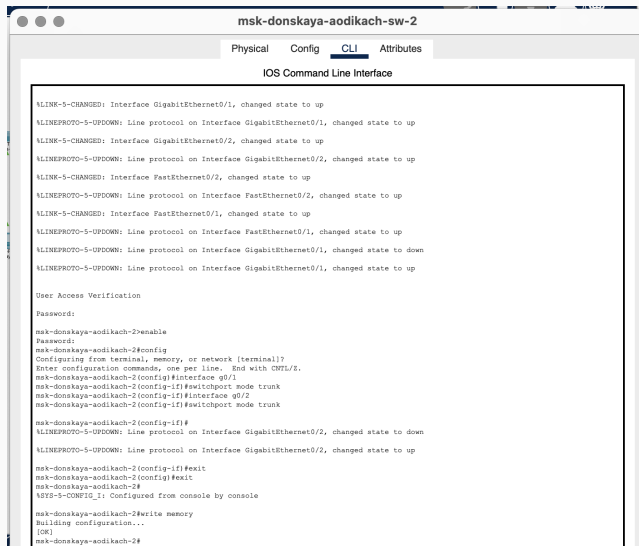
msk-donskaya-aodikach-sw-1#
VCDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (101), with msk-pavlovskaya-aodikach-sw-1
FastEthernet0/24 (1).

msk-donskaya-aodikach-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-aodikach-sw-1#

```

Рис. 4: Настройка коммутаторов

Провожу настройку конфигурации Trunk-портов, VTP и диапазона портов для msk-donskaya-aodikach-sw-02



```
msk-donskaya-aodikach-sw-2
Physical  Config  CLI  Attributes
IOS Command Line Interface

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

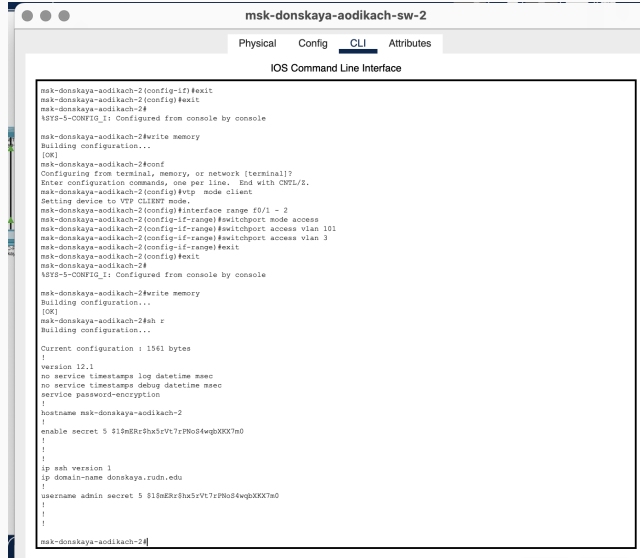
User Access Verification

Password:
msk-donskaya-aodikach-2>enable
Password:
msk-donskaya-aodikach-2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-2(config)#interface g0/1
msk-donskaya-aodikach-2(config-if)#switchport mode trunk
msk-donskaya-aodikach-2(config-if)#interface g0/2
msk-donskaya-aodikach-2(config-if)#switchport mode trunk

msk-donskaya-aodikach-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

msk-donskaya-aodikach-2(config-if)#exit
msk-donskaya-aodikach-2(config)#exit
msk-donskaya-aodikach-2#
SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aodikach-2#write memory
Building configuration...
[OK]
msk-donskaya-aodikach-2#
```



The screenshot shows a network device configuration window titled "msk-donskaya-aodikach-sw-2". It has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" being the active tab. The main area is titled "IOS Command Line Interface" and contains a list of configuration commands and their outputs. The commands are entered in a terminal-like environment, and the outputs show the current configuration state, including VTP mode, interface range configuration, and the current configuration text.

```
msk-donskaya-aodikach-2(config-if)#exit
msk-donskaya-aodikach-2(config)#exit
msk-donskaya-aodikach-2#
$SYS-5-CONFIG_I: Configured from console by console

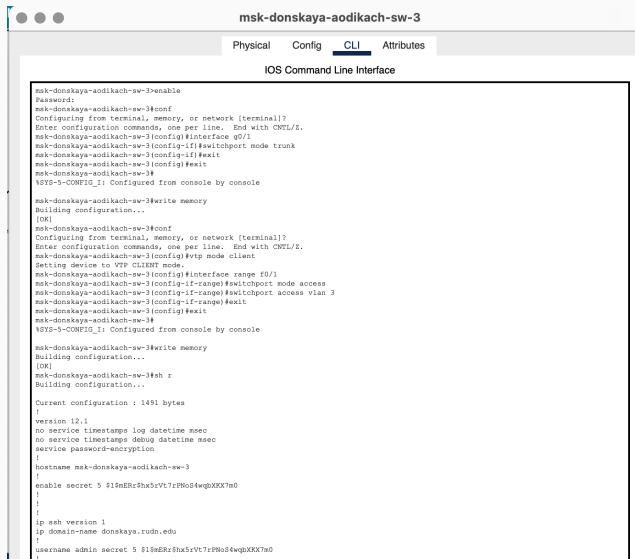
msk-donskaya-aodikach-2#write memory
Building configuration...
[OK]
msk-donskaya-aodikach-2#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-2(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-aodikach-2(config)#interface range f0/1 - 2
msk-donskaya-aodikach-2(config-if-range)#switchport mode access
msk-donskaya-aodikach-2(config-if-range)#switchport access vlan 101
msk-donskaya-aodikach-2(config-if-range)#switchport access vlan 3
msk-donskaya-aodikach-2(config-if-range)#exit
msk-donskaya-aodikach-2(config)#exit
msk-donskaya-aodikach-2#
$SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aodikach-2#write memory
Building configuration...
[OK]
msk-donskaya-aodikach-2#sh r
Building configuration...

Current configuration : 1561 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname msk-donskaya-aodikach-2
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
ip ssh version 1
ip domain-name donskeya.rudn.edu
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
msk-donskaya-aodikach-2#
```

Рис. 6: Настройка VTP и диапазона портов

Провожу настройку конфигурации Trunk-портов, VTP и диапазона портов для msk-donskaya-aodikach-sw-03



```
msk-donskaya-aodikach-sw-3
Physical  Config  CLI  Attributes

IOS Command Line Interface

msk-donskaya-aodikach-sw-3>enable
Password:
msk-donskaya-aodikach-sw-3#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-sw-3(config)#interface g0/1
msk-donskaya-aodikach-sw-3(config-if)#switchport mode trunk
msk-donskaya-aodikach-sw-3(config-if)#exit
msk-donskaya-aodikach-sw-3(config)#exit
msk-donskaya-aodikach-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aodikach-sw-3#write memory
Building configuration...
[OK]
msk-donskaya-aodikach-sw-3#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-sw-3(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-aodikach-sw-3(config)#interface range f0/1
msk-donskaya-aodikach-sw-3(config-if-range)#switchport mode access
msk-donskaya-aodikach-sw-3(config-if-range)#switchport access vlan 3
msk-donskaya-aodikach-sw-3(config-if-range)#exit
msk-donskaya-aodikach-sw-3(config)#exit
msk-donskaya-aodikach-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aodikach-sw-3#write memory
Building configuration...
[OK]
msk-donskaya-aodikach-sw-3#sh r
Building configuration...

Current configuration : 1491 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname msk-donskaya-aodikach-sw-3
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXXK7m0
!
!
!
ip ssh version 1
ip domain-name donskeya.rudn.edu
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXXK7m0
```

Провожу настройку конфигурации Trunk-портов, VTP и диапазона портов для msk-donskaya-aodikach-sw-04

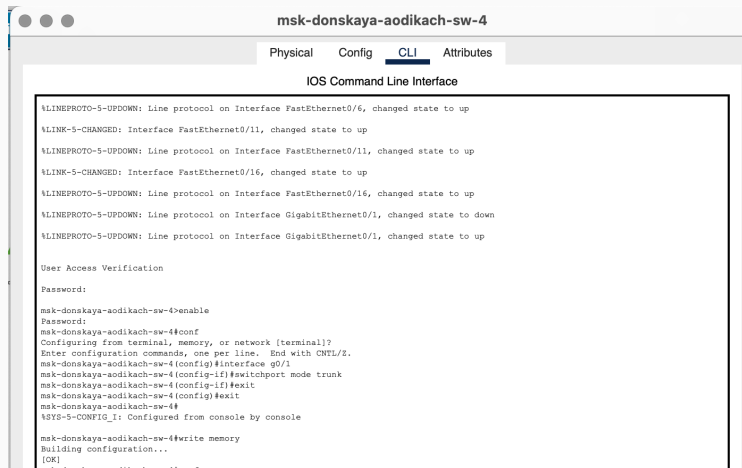


Рис. 8: Настройка коммутатора

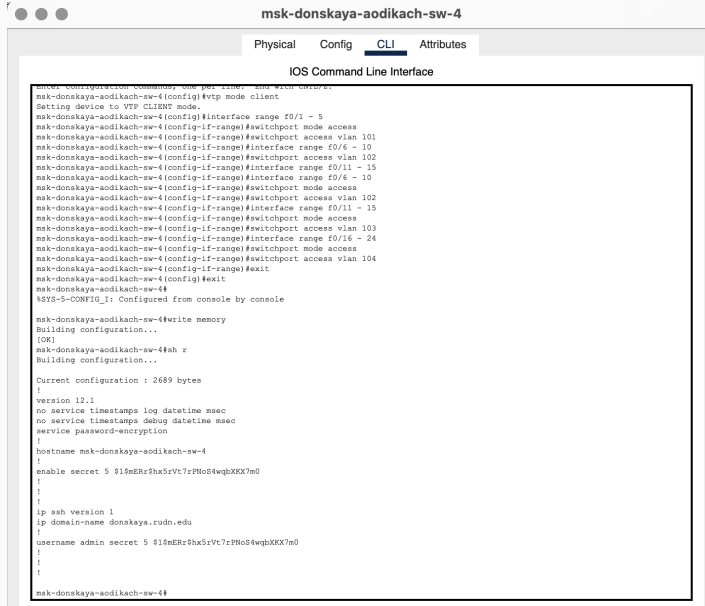


Рис. 9: Настройка коммутатора

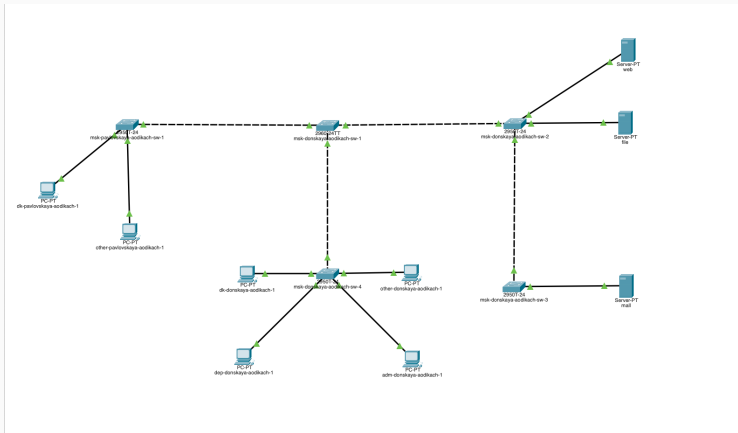


Рис. 10: Построенная и настроенная сеть

Проверяю результат проделанной работы на der-donskaya-aodikach-1 и пробую пропинговать устройство из другой сети. Неудачно



Рис. 11: Проверка корректности

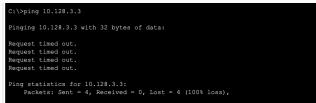


Рис. 12: Неудачная попытка

Запускаю симуляцию. В сети на одной улице пакеты коммуницируют друг с другом верно, однако при попытке передать пакеты на другую улицу вылезает ошибка

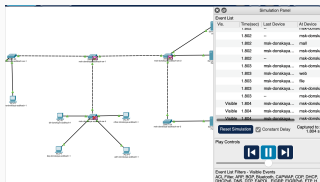


Рис. 13: Запуск симуляции

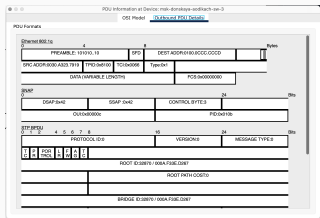


Рис. 14: Разбор одного из отправленных пакетов

Получила основные навыки по настройке VLAN на коммутаторах сети.